Title

Hierarchical Clustering Analysis on Customer Segmentation Dataset

Introduction

Clustering is a technique in data analysis that groups similar items together, making it easier to uncover patterns and relationships within a dataset. One widely used approach is Hierarchical Clustering, which gradually forms groups in a step-by-step manner and represents them in a tree-like structure known as a dendrogram. Unlike some other clustering methods, hierarchical clustering does not require specifying the number of clusters beforehand, making it especially valuable when the natural groupings in the data are unknown.

The goal of this experiment is to understand the working process of hierarchical clustering, learn how to interpret dendrograms, and explore how clustering can help reveal hidden patterns and meaningful structures in customer data.

Required Library

- Pandas
- Scikit-learn
- Matplotlib
- Seaborn
- Scipy
- Seaborn

Theory

Clustering is a data analysis technique that groups data points in such a way that objects within the same group are more similar to each other than to those in different groups.Hierarchical clustering is a special type of clustering that builds a hierarchy of clusters, represented in a tree-like structure known as a dendrogram.There are two main approaches to hierarchical clustering:

1. Top-Down Clustering (Divisive Method)

In the divisive approach, the algorithm begins with all data points combined into a single, large cluster. This cluster is then split into smaller clusters based on dissimilarities among the data points. The process of division continues until each point forms its own cluster, or until the required number of clusters is achieved. This method can be described as a step-by-step splitting process.

2. Bottom-Up Clustering (Agglomerative Method)

In the agglomerative approach, the process starts with each data point treated as its own individual cluster. At each step, the two clusters that are most similar, according to a chosen distance measure (e.g., Euclidean distance), are merged together. This merging process continues until all points are grouped into a single cluster or until the desired number of clusters is reached. This method can be described as a step-by-step merging process.

Code

Code	Explanation

`import pandas as pd`
`import matplotlib.pyplot as plt`
`import seaborn as sns`
`from sklearn.preprocessing import LabelEncoder, StandardScaler`
`from scipy.cluster.hierarchy import dendrogram, linkage`
`from sklearn.cluster import AgglomerativeClustering`
	In this step, the necessary Python libraries are imported: Pandas for data handling, Matplotlib and Seaborn for visualization, and Scikit-learn for preprocessing and clustering. SciPy functions are also imported for hierarchical clustering, and the plot style is set with a white Seaborn background for clear, readable visualizations.

`df = pd.read_csv('/content/PRL_12_8_2025.csv')`
`display(df.head())`
 	
`print(df.columns)`
`df.info()`
	The dataset is loaded from the CSV file into a Pandas DataFrame. The first few rows, along with column names and data types, are displayed to verify the data and identify categorical and numerical features for clustering.

`fig, axes = plt.subplots(2, 2, figsize=(12, 8))`
`axes[0, 0].hist(df['Age'], bins=15, color='#0B3D91', edgecolor='black')`
`axes[0, 0].set_title('Age Distribution', color='black')`
`axes[0, 1].hist(df['Annual Income (k$)'], bins=15, color='#8B0000', edgecolor='black')`
`axes[0, 1].set_title('Annual Income Distribution', color='black')`
`axes[1, 0].hist(df['Spending Score (1-100)'], bins=15, color='#556B2F', edgecolor='black')`
`axes[1, 0].set_title('Spending Score Distribution', color='black')`
`axes[1, 1].bar(df['Genre'].value_counts().index,`
`df['Genre'].value_counts().values,`
`color=['pink', 'skyblue'])`
`axes[1, 1].set_title('Genre Count', color='black')`
`plt.tight_layout()`
`plt.show()`
	At this stage, the dataset is explored visually. Histograms are plotted for the Age, Annual Income, and Spending Score attributes, each represented with dark colors for better contrast. Additionally, a bar chart is drawn to show the distribution of the Genre variable. These visualizations provide insights into the spread and frequency distribution of customer characteristics, which is useful for detecting patterns before applying clustering methods.

Code	Explanation

`label_encoder = LabelEncoder()`
`df['Genre_Encoded'] = label_encoder.fit_transform(df['Genre'])`
`display(df.head())`
	The categorical column Genre is converted into numeric form using Label Encoding, creating a new column Genre_Encoded so the dataset becomes compatible with clustering algorithms.

`X = df[['Annual Income (k$)', 'Spending Score (1-100)']]`
`linked = linkage(X, method='ward')`
`dark_colors = ['#0B3D91', '#8B0000', '#556B2F', '#4B0082', '#FF8C00']`
`plt.figure(figsize=(12, 7))`
`dendrogram(linked,`
`orientation='top',`
`distance_sort='descending',`
`show_leaf_counts=True,`
`color_threshold=7,`
`above_threshold_color='gray',`
`link_color_func=lambda k:dark_colors[k % len(dark_colors)])`
`plt.title('Dendrogram (Unscaled Data)', color='black')`
`plt.xlabel('Customers', color='black')`
`plt.ylabel('Distance', color='black')`
`plt.show()`
	Hierarchical clustering is first applied to the unscaled dataset using Ward's method. A dendrogram is generated to visualize the hierarchical relationships among data points. To make the dendrogram more visually appealing, a custom dark color palette is applied, producing multiple dark-colored branches. This allows us to see potential groupings in the customer data without any scaling applied.

`n_clusters = 5`
`agg_clustering = AgglomerativeClustering(n_clusters=n_clusters)`
`df['Cluster_Unscaled'] = agg_clustering.fit_predict(X)`
`plt.figure(figsize=(10, 6))`
`sns.scatterplot(x='Annual Income (k$)', y='Spending Score (1-100)',`
`hue='Cluster_Unscaled', data=df,`
`palette='coolwarm', s=100)`
`plt.title('Clusters on Unscaled Data', color='black')`
`plt.xlabel('Annual Income (k$)', color='black')`
`plt.ylabel('Spending Score (1-100)', color='black')`
`plt.grid(True)`
`plt.show()`
	Using the dendrogram as a guide, Agglomerative Clustering is performed with five clusters on the unscaled features Annual Income and Spending Score. The results are visualized in a scatter plot, where each cluster is shown in a distinct color using the coolwarm palette. This plot helps in observing how customers are segmented into different groups based on their income and spending behavior before applying scaling.

Code	Explanation

`scaler = StandardScaler()`
`df[['Annual Income (k$)', 'Spending Score (1-100)']] = scaler.fit_transform(df[['Annual Income (k$)', 'Spending Score (1-100)']])`
`display(df.head())`
	The numerical features are standardized using StandardScaler to have zero mean and unit variance. This ensures that features with different ranges don't dominate clustering. The scaled dataset is then displayed to confirm normalization.

`X_scaled = df[['Annual Income (k$)', 'Spending Score (1-100)']]`
`linked_scaled = linkage(X_scaled, method='ward')`
`plt.figure(figsize=(12, 7))`
`dendrogram(linked_scaled,`
`orientation='top',`
`distance_sort='descending',`
`show_leaf_counts=True,`
`color_threshold=3,`
`above_threshold_color='gray',`
`link_color_func=lambda k: dark_colors[k % len(dark_colors)])`
`plt.title('Dendrogram (Scaled Data)', color='black')`
`plt.xlabel('Samples', color='black')`
`plt.ylabel('Distance', color='black')`
`plt.show()`
	Hierarchical clustering is repeated on the scaled dataset. A new dendrogram is generated with the same dark multi-color palette to visualize the new structure of clusters. Because the data has been scaled, the distances are now more balanced, leading to more accurate hierarchical groupings. The use of a threshold ensures that clusters are visually separated with distinct colors.

`agg_clustering_scaled = AgglomerativeClustering(n_clusters=n_clusters)`
`df['Cluster_Scaled'] = agg_clustering_scaled.fit_predict(X_scaled)`
 	
`plt.figure(figsize=(10, 6))`
`sns.scatterplot(x='Annual Income (k$)', y='Spending Score (1-100)',`
`hue='Cluster_Scaled', data=df,`
`palette='Set2', s=100)`
`plt.title('Clusters on Scaled Data', color='black')`
`plt.xlabel('Annual Income (k$)', color='black')`
`plt.ylabel('Spending Score (1-100)', color='black')`
`plt.grid(True)`
`plt.show()`
	Finally, Agglomerative Clustering is applied to the scaled dataset with the same number of clusters. The results are shown in a scatter plot using the Set2 palette, which provides a different visual style from the previous clustering plot. This comparison highlights the difference between clustering with raw data versus standardized data, demonstrating the importance of scaling in unsupervised learning tasks.

Output

	CustomerID	Genre	Age	Annual Income (k$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40

Figure 1: Display The Data

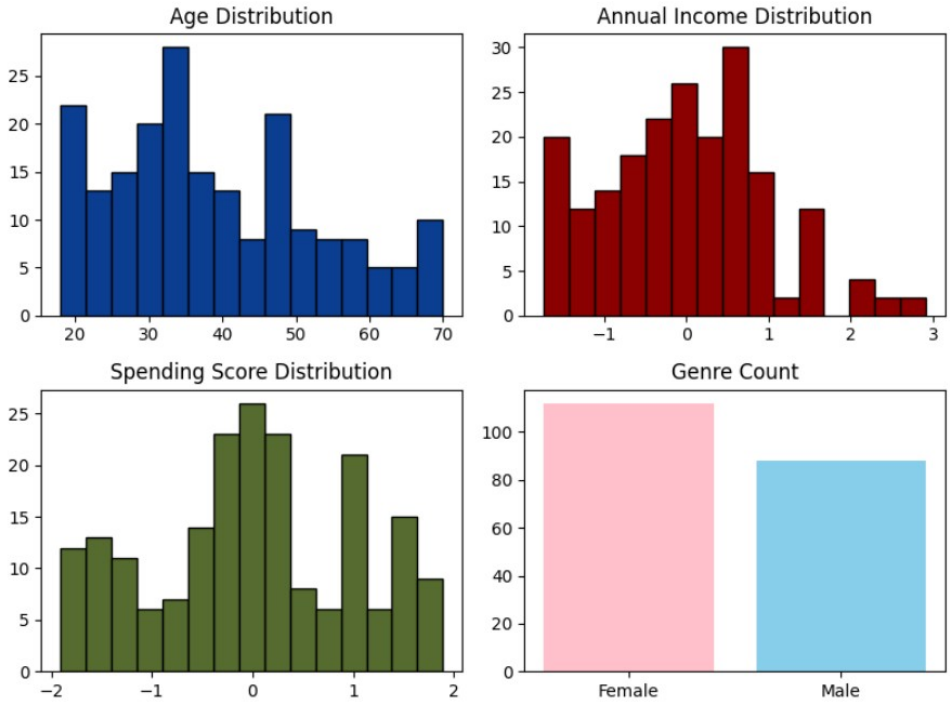

Figure 2: Subplot of Data

	CustomerID	Genre	Age	Annual Income (k$)	Spending Score (1-100)	Genre_Encoded
0	1	Male	19	15	39	1
1	2	Male	21	15	81	1
2	3	Female	20	16	6	0
3	4	Female	23	16	77	0
4	5	Female	31	17	40	0

Figure 3: Genre_Encoded

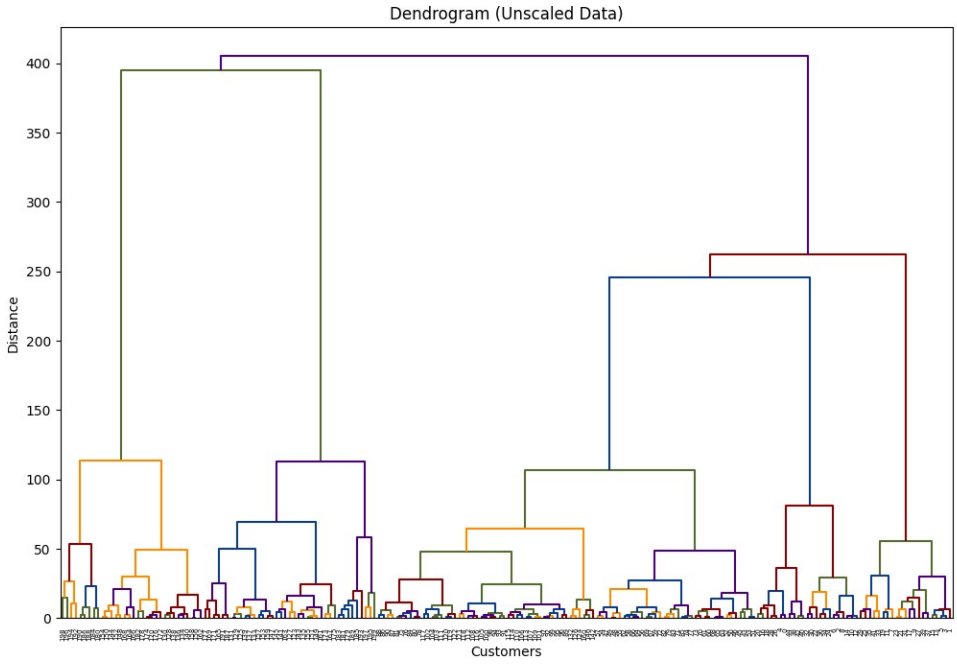

Figure 4: Dendrogram (Unscaled Data)

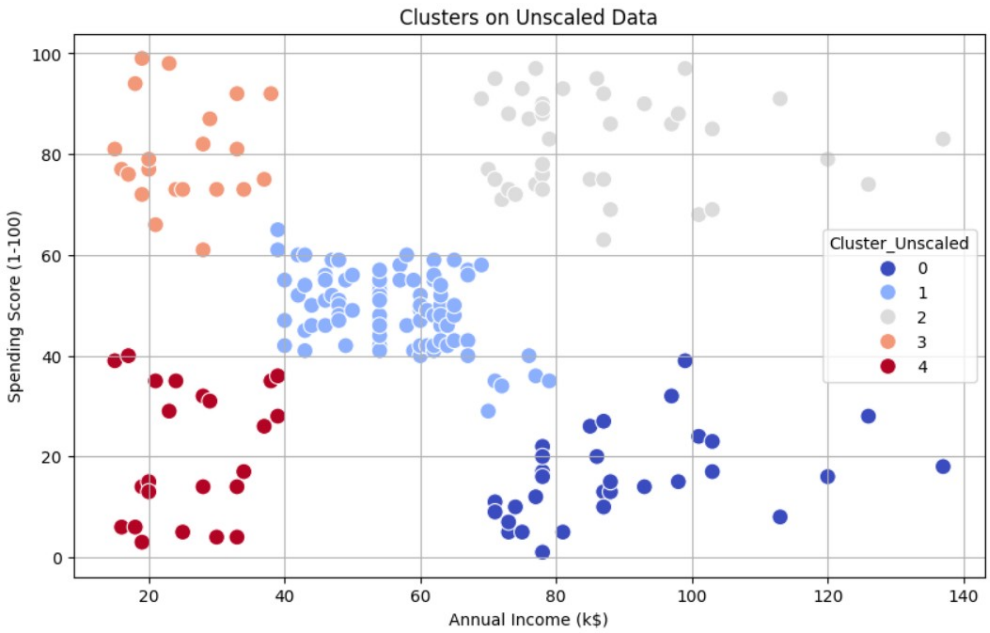

Figure 5: Clusters on Unscaled Data

	CustomerID	Genre	Age	Annual Income (k$)	Spending Score (1-100)	Genre_Encoded	Cluster_Unscaled
0	1	Male	19	-1.738999	-0.434801	1	4
1	2	Male	21	-1.738999	1.195704	1	3
2	3	Female	20	-1.700830	-1.715913	0	4
3	4	Female	23	-1.700830	1.040418	0	3
4	5	Female	31	-1.662660	-0.395980	0	4

Figure 6: Clustered Customer Data

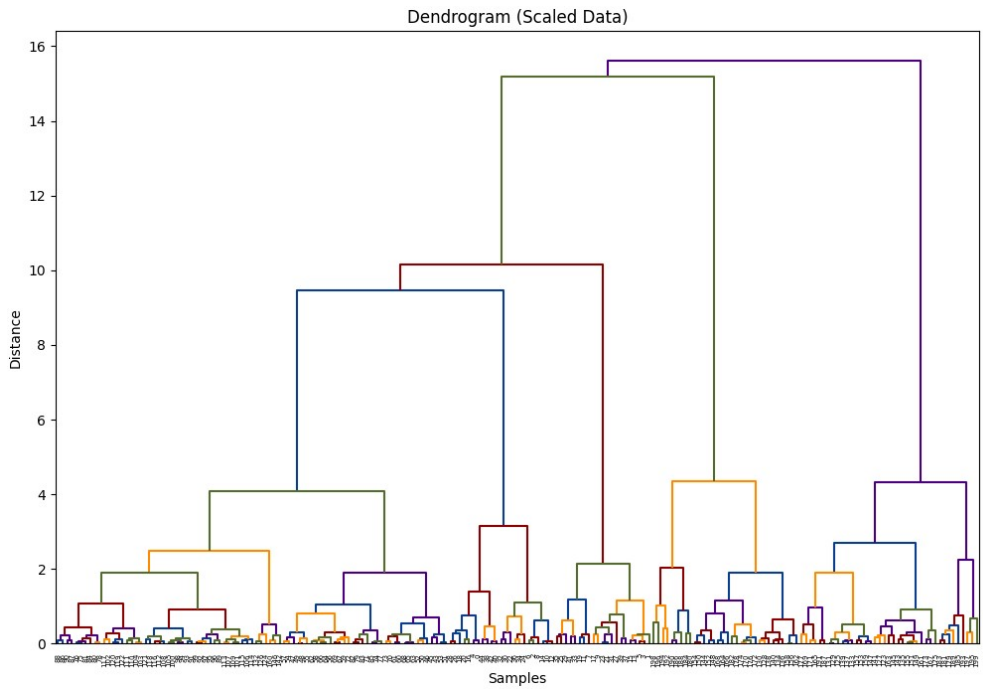

Figure 7: Dendrogram (Scaled Data)

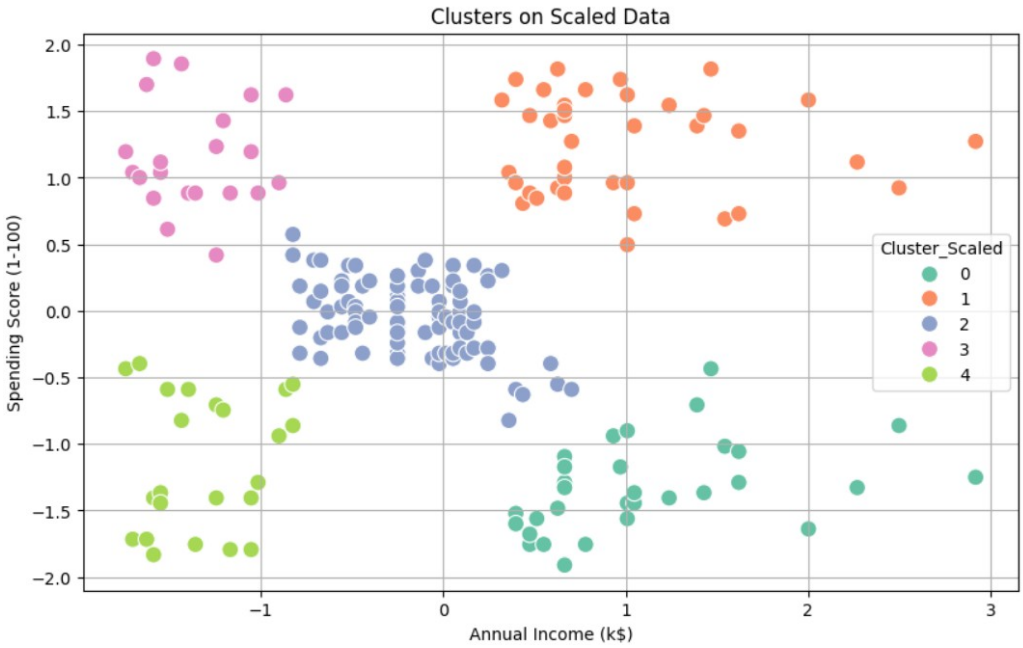

Figure 8: Clusters on Scaled Data

Discussion

Hierarchical clustering was applied to group customers based on Annual Income and Spending Score. While clustering on unscaled data revealed general trends, scaling with StandardScaler improved accuracy by giving both features equal importance. Dendrograms helped determine the optimal number of clusters, and scatter plots visualized the distinct customer segments. The results show that hierarchical clustering effectively identifies patterns and groups similar customers, which can aid in targeted marketing strategies.